rLEDBAT Modification Study

Final Findings Report

CADF · ECS · ATD | ns-3 Simulation | 10 Mbps / 20 ms Bottleneck | Target Delay 5 ms

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1. Introduction

The Internet carries a heterogeneous mix of traffic classes. Latency-sensitive flows — video calls, interactive gaming, and remote desktop sessions — compete for bandwidth with background transfers such as software update distribution and cloud backup. When background traffic is managed by a standard congestion control algorithm such as CUBIC or NewReno, it consumes bandwidth aggressively, inflating queuing delay and degrading the quality of interactive applications sharing the same bottleneck.

LEDBAT (Low Extra Delay Background Transport, RFC 6817) is a delay-based, less-than-best-effort (LBE) algorithm that reduces its sending rate when measured queuing delay exceeds a configured target, yielding capacity to competing traffic. LEDBAT++ refines this with RTT-based measurement, additive increase / multiplicative decrease (AI/MD) for inter-flow fairness, and periodic slowdowns to refresh the base delay estimate.

rLEDBAT (receiver-driven LEDBAT) implements the LBE controller at the TCP receiver, communicating the desired rate to the sender by manipulating the TCP receive window (RCV.WND). The sender computes its sending window as min(cwnd, RCV.WND); rLEDBAT keeps RCV.WND below standard congestion control limits, transparently throttling the sender without any sender-side modifications.

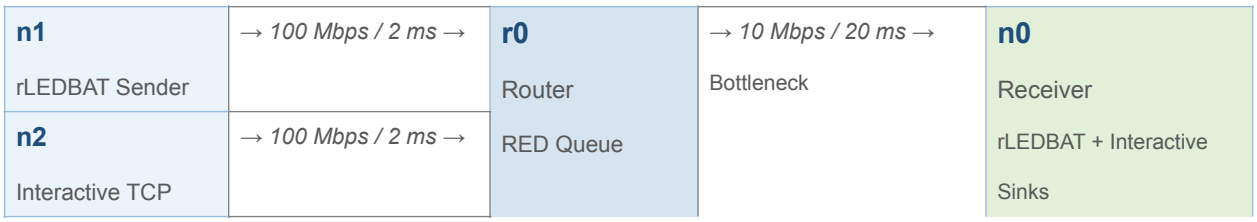
This report documents an ns-3 implementation of rLEDBAT and evaluates three receiver-side modifications:

- **CADF — Channel-Aware Delay Filtering:** suppresses W_ledbat updates when a qdelay sample is a transient path-jitter spike, preventing unnecessary window reductions without masking genuine congestion.
- **ECS — Explicit Congestion Signalling:** reacts to ECN CE marks in the IP header via a ForwardUp() override, applying a multiplicative decrease before the delay signal reaches the target threshold.
- **ATD — Adaptive Target Delay:** computes effective_target = max(base_delay × alpha, atdMinTarget), scaling the target with the measured path RTT to avoid manual per-topology tuning.

2. Network Topology and Simulation Setup

2.1 Topology

The simulation uses a five-node dumbbell topology implemented with ns-3 PointToPointHelper. Three senders converge at a shared router (r0) before traversing a shared bottleneck link to the receiver (n0).



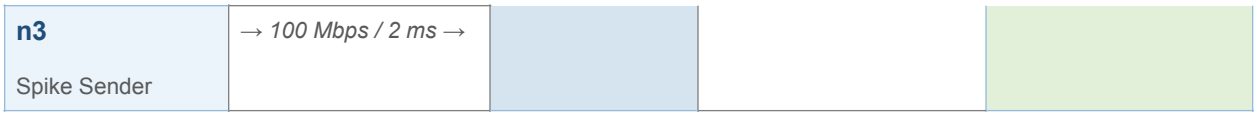


Figure 1: Five-node dumbbell topology. Access links (100 Mbps / 2 ms), bottleneck r0→n0 (10 Mbps / 20 ms). Total propagation RTT = 2×(2+20) = 44 ms.

Node roles: n0 = receiver (rLEDBAT sink on port 9000, interactive sink on port 9001). n1 = rLEDBAT background sender. n2 = interactive high-priority sender. n3 = delay-spike cross-traffic sender. r0 = aggregation router with RED queue disc.

Access links (n1/n2/n3 → r0): 100 Mbps, 2 ms propagation, 500-packet DropTail queue at device level.

Bottleneck link (r0 → n0): 10 Mbps, 20 ms propagation. A 4-packet device-level queue forces all buffering through the traffic-control queue disc.

Queue disc — baseline (ECN=Off): RED with MinTh=150p, MaxTh=185p, MaxSize=200p. With MinTh near queue capacity, RED behaves as a large DropTail. rLEDBAT sees queuing delay build freely and self-throttles via its delay signal.

Queue disc — ECS runs (ECN=On): RED with MinTh=3p (~3.5 ms queuing), MaxTh=100p, MaxSize=200p. CE marking begins well before the rLEDBAT target delay, giving ECS its early-warning advantage.

The rLEDBAT sink on n0 is implemented via a CustomSink application class that temporarily overrides TcpL4Protocol::SocketClass during socket creation so that connections on port 9000 use TcpRledbat. The interactive sink on port 9001 uses TcpSocketBase. All senders use BulkSendHelper (unlimited data, stopped by time). TCP congestion control at all senders is TcpNewReno with timestamps enabled globally.

2.2 Test Combinations

Each simulation run covers three sequential phases separated by 2-second draining gaps:

Phase	Time (s)	Active Flows	Purpose
Combo 1	1 – 20	n1 (rLEDBAT) alone	Baseline: rLEDBAT fills bottleneck alone
Spike 1	8 – 9	n3 burst (queue congestion)	Fills bottleneck queue; tests window recovery
Spike 2	11 – 11.04	n1↔r0 delay raised 2→40 ms	Path jitter artifact; queue stays empty; tests CADF
Combo 2	22 – 40	n2 (interactive) alone	Reference: unconstrained TCP fills 10 Mbps link
Combo 3	42 – 62	n1 + n2 simultaneously	Yielding: rLEDBAT must back off for interactive flow

Table 1: Simulation phase structure

3. rLEDBAT Core Algorithm Implementation

RTT estimation: TSecr-based RTT = now − MilliSeconds(TSecr). TSecr deduplication skips repeated TSecr values. A min-of-K=4 rolling filter produces the filtered RTT used as the current delay (sec. 4.2.1).

Base delay: Epoch-minimum tracking over a 5-second refresh interval, history depth 2. Base delay = minimum filtered RTT seen in the current epoch or any stored epoch minimum.

Queuing delay: qd = max(0, currentDelay − baseDelay). When qd = 0 the path is uncongested; rLEDBAT increases the window.

Window update (RFC 6817): offTarget = (TARGET − qd) / TARGET; adj = GAIN × offTarget × ackedBytes × MSS / RLWND. ackedBytes is accumulated before UpdateRledbatWindow() is called per packet.

Retransmission detection (sec. 4.3): If SEG.SEQ < RCV.HGH and TSV.SEQ > TSV.HGH, the packet is a retransmission. Multiplicative decrease (RLWND ÷ 2) applied once per RTT.

Anti-shrink (sec. 4.1.1): Advertised RCV.WND may only decrease by at most ackedBytes per outgoing ACK, preventing RFC 9293-prohibited window shrinking.

4. Modifications to the Simulator

4.1 Mod 1: Channel-Aware Delay Filtering (CADF)

Without filtering, a single-RTT path jitter event (route change, wireless channel variation) causes rLEDBAT to reduce its window unnecessarily, degrading throughput. CADF extends UpdateRledbatWindow() with a two-condition spike detector and a streak-based classifier.

Spike detection: A qdelay sample is a spike candidate if (1) qd > CadfSpikeRatio × mean(history) AND (2) qd − mean > CadfMinAbsoluteSpike. Both conditions must hold. Condition (2) prevents false positives when the baseline qdelay is near zero: e.g. 0.5 ms mean → 1.5 ms sample passes the 2.5× ratio but is only 1 ms of TCP jitter, not a real spike.

History update: All qdelay samples — including below-threshold spikes — are added to the CADF history deque (window size 8). The CADF history tracks the recent delay distribution for spike classification; excluding spike samples would prevent the mean from adapting during genuine sustained congestion. Only the streak counter is not updated for non-spike samples.

Streak-based classification: Each spike increments the streak counter. If streak < CadfStreakThreshold (4): the window update is suppressed, but the sample IS added to history. Once streak ≥ 4, the

elevated delay is persistent congestion — the window update proceeds. Non-spike samples reset the streak to 0.

Parameter	Value	Rationale
CadfStreakThreshold	4	Spike 2 lasts ≈50 ms (1 RTT → 1–2 spike packets); streak stays below 4 → suppressed
CadfWindowSize	8	≈2× threshold; mean adapts within 2 streak lengths of persistent congestion
CadfMinAbsoluteSpike	5 ms	Matches TargetDelay default; sub-5 ms jitter is not classified as a spike
CadfSpikeRatio	2.5	Sample must exceed 2.5× recent mean to qualify as a spike candidate

Table 2: CADF parameter values and rationale

4.2 Mod 2: Explicit Congestion Signalling (ECS)

Standard rLEDBAT reacts to congestion only after measuring elevated queuing delay, requiring at least one RTT of buffering. ECS adds a parallel congestion signal: IP-layer ECN Congestion Experienced (CE) marking. When the RED queue disc is configured with ECN=On and MinTh=3p (≈3.5 ms queuing), the router marks packets before the delay-based controller detects the queue build.

ECS intercepts CE marks in TcpRledbat::ForwardUp() — before the base class processes the IP header — and applies an immediate multiplicative window decrease: RLWND ← max(RLWND × EcsBeta, MinWindow), gated once per RTT using a time-based guard (m_lastEcsDecreaseTime). The time gate prevents cascading reductions from back-to-back CE-marked packets.

Important note on ECN scope: the ECS modification sets Config::SetDefault("ns3::TcpSocketBase::UseEcn", StringValue("On")) globally when ECN is enabled. This applies to ALL TCP sockets created in the simulation — including the BulkSend senders for both n1 and n2, and both sinks. Standard TCP's ECE/CWR mechanism therefore also constrains the interactive flow (n2). This is confirmed by the data: Combo 2 interactive throughput drops from 9.81 Mbps (base) to 8.10 Mbps (ECS), even though n1 is completely absent during Combo 2. Any interpretation of ECS Combo 3 results must account for this global ECN effect on both flows.

Parameter	Values Tested	Effect
EcsBeta	0.2, 0.5, 0.8	Multiplicative decrease factor per CE mark. Lower = more aggressive reduction per event.
MinTh (RED, ECN)	3 pkts (≈3.5 ms)	CE marking begins before rLEDBAT target delay; ECS fires with ≈1 RTT early warning.
MaxTh (RED, ECN)	100 pkts	Wide probabilistic marking range; reduces abrupt packet drops.

Parameter	Values Tested	Effect
MinTh (RED, no ECN)	150 pkts	Near queue capacity; RED acts as DropTail backstop only, preserving delay signal.

Table 3: ECS parameters and queue configuration

4.3 Mod 3: Adaptive Target Delay (ATD)

rLEDBAT's fixed TargetDelay creates a mismatch in heterogeneous networks where the inherent propagation delay changes across paths. If the base RTT is much larger than the target, rLEDBAT permanently perceives queuing delay and collapses its window. ATD replaces the fixed target with a dynamically scaled value:

effectiveTarget = max(baseDelay × AtdAlpha, AtdMinTarget)

When baseDelay is large (long-path or mobile scenarios), effectiveTarget scales proportionally, keeping the operating point as a fixed fraction of the propagation RTT. AtdMinTarget (default 0 ms) ensures the target never collapses to zero on very short paths.

In this wired simulation the base delay is stable at 44 ms throughout. ATD with alpha=0.5 produces effectiveTarget = max(0.5×44, 0) = 22 ms — substantially higher than the fixed 0 ms default. This permits rLEDBAT to maintain a larger window in Combo 1, driving higher throughput, while still yielding to interactive traffic in Combo 3.

Parameter	Values Tested	Effect
AtdAlpha	0.2, 0.5, 0.8	Scaling factor. Higher alpha → larger effective target → less throttling → higher rLEDBAT throughput.
AtdMinTarget	5 ms	Floor on effective target; prevents collapse on very short propagation paths.

Table 4: ATD parameters

5. Results

5.1 Base rLEDBAT: Target Delay Sweep

Three values of TargetDelay were evaluated with all modifications disabled. Figure 2 shows the rLEDBAT internal state trace for the default 5 ms target. The base delay statistics across the full simulation are: min=44 ms, max=48 ms, mean=44.18 ms — confirming the base delay remains nearly constant at the propagation RTT of 44 ms throughout all combinations, with only a negligible transient rise.

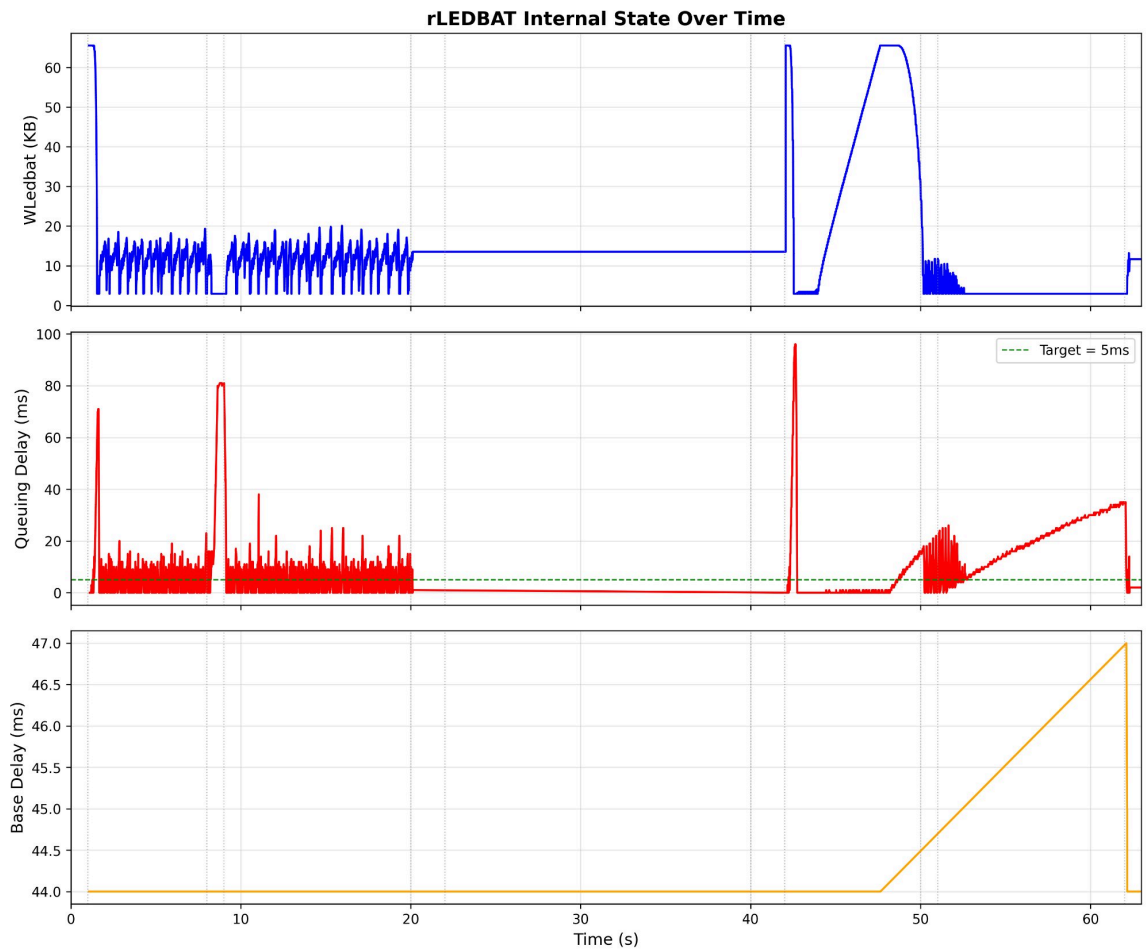


Figure 2: Base rLEDBAT (target=5 ms). Top: W_ledbat oscillates continuously between ~3 KB and 65 KB in Combo 1 — the 5 ms target is tight enough that the window is frequently pulled down whenever the queue builds. Spikes 1 (t=8s) and 2 (t=11s) cause deeper drops. In Combo 3 the window crashes to near-zero and shows sawtooth cycles. Middle: queuing delay spikes at startup, t=8s (~70 ms), t=11s (~60 ms), and persistently above target in Combo 3. Bottom: base delay flat at 44 ms throughout.

The window statistics for the 5 ms base run confirm the oscillatory nature: mean W_ledbat = 20.54 KB (not the 65 KB maximum), min = 2.92 KB. The algorithm is continuously regulating the window around the target — it does not ramp to the maximum and stay there. In Combo 3, the window collapses because the interactive flow saturates the bottleneck, pushing qd far above the 5 ms target. Each cycle in the Combo 3 sawtooth represents rLEDBAT attempting recovery before being pushed down again by sustained competition.

Target	C1: rLEDBAT (Mbps)	C1: E2E Delay (ms)	C1: PDR (%)	C2: Interactive (Mbps)	C3: rLEDBAT (Mbps)	C3: Interactive (Mbps)	C3: rLEDBAT PDR (%)	C3: Interactive PDR (%)	C3 Jain	Net Tput (Mbps)	Net PDR (%)
2.5 ms	5.16	26	100.00	9.81	1.21	8.53	99.92	99.98	0.639	7.63	99.99
5 ms	7.30	30	100.00	9.81	2.32	6.05	99.56	99.81	0.834	7.83	99.91

Target	C1: rLEDBAT (Mbps)	C1: E2E Delay (ms)	C1: PDR (%)	C2: Interactive (Mbps)	C3: rLEDBAT (Mbps)	C3: Interactive (Mbps)	C3: rLEDBAT PDR (%)	C3: Interactive PDR (%)	C3 Jain	Net Tput (Mbps)	Net PDR (%)
15 ms	8.81	39	100.00	9.81	2.75	5.71	99.35	99.70	0.890	8.27	99.86

Table 5: Base rLEDBAT — TargetDelay sweep. Per-combination and network-wide metrics.

Lower targets produce stronger yielding (rLEDBAT more throttled, interactive gets more) but at the cost of reduced solo throughput in Combo 1. At 2.5 ms, rLEDBAT achieves only 5.16 Mbps alone — the tight target prevents the window from growing large enough to saturate the link. At 15 ms, rLEDBAT reaches 8.81 Mbps alone but yields less aggressively in Combo 3 (5.71 Mbps interactive vs 9.81 Mbps when alone). The 5 ms target provides a practical balance. Drop ratios are near-zero for Combo 1 at all targets; Combo 3 drops increase with target (0.08% at 5 ms for rLEDBAT) because a larger window means more in-flight packets when retransmissions occur.

Node / Metric	Target=2.5ms	Target=5ms	Target=15ms
n1 throughput (Mbps, 39s)	3.18	4.79	5.74
n1 PDR (%)	99.98	99.89	99.84
n1 drop ratio (%)	0.019	0.111	0.161
n2 throughput (Mbps, 38s)	9.18	7.87	7.69
n2 PDR (%)	99.99	99.92	99.88
n3 (spike, 1s) throughput (Mbps)	7.87	7.59	4.45
n3 PDR (%)	100.00	100.00	99.27
Network total throughput (Mbps)	7.63	7.83	8.27
Network PDR (%)	99.99	99.91	99.86
Network drop ratio (%)	0.015	0.090	0.143

Table 6: Base rLEDBAT — per-node and network-wide aggregate metrics (all combos combined)

5.2 CADF: Channel-Aware Delay Filtering

CADF was evaluated with CadfStreakThreshold=4, CadfWindowSize=8, CadfMinAbsoluteSpike=5 ms, and TargetDelay=5 ms. Figure 3 shows the internal state trace. As expected, the CADF trace is essentially identical to the base trace at this time scale: the aggregate metrics are nearly unchanged, and the W_ledbat, qdelay, and base delay patterns are visually indistinguishable from Figure 2.

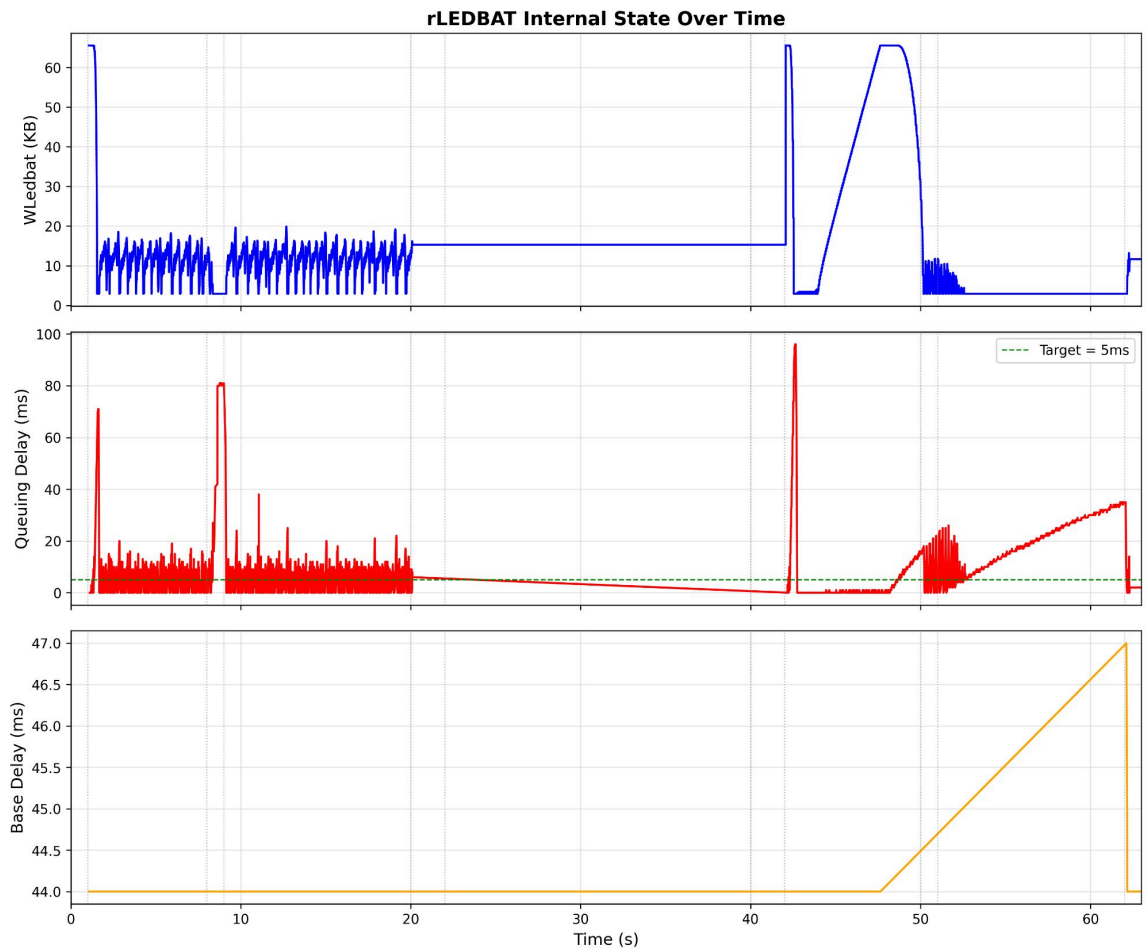


Figure 3: CADF mode. The trace is nearly identical to the base (Figure 2): W_ledbat oscillates continuously throughout Combo 1 with the same amplitude and spike responses. Spike 2 (t=11s) protection — 1–2 packets with streak < 4 — is invisible at this scale. Combo 3 sawtooth behavior is the same as base. Base delay flat at 44 ms.

The near-identical appearance is expected and correct. Spike 2 at t=11–11.04s lasts 40 ms, which at the 44 ms RTT generates at most 1–2 consecutive spike packets (streak ≤ 2 < threshold of 4). Those 1–2 window updates are suppressed, which is undetectable in flow-level metrics. Spike 1 at t=8–9s lasts 1 second (≈22 RTTs); after 4 consecutive spike packets (≈176 ms), the streak threshold is crossed and CADF correctly allows the window update to proceed, yielding the same Combo 3 behaviour as base. The CADF history (window size 8) always receives all samples — spike or not — so the mean adapts during genuine congestion without being frozen at a stale value.

Metric	Base (5 ms)	CADF
C1: rLEDBAT throughput (Mbps)	7.30	7.27
C1: E2E delay (ms)	30	30
C1: PDR (%)	100.00	100.00
C2: Interactive throughput (Mbps)	9.81	9.81
C3: rLEDBAT throughput (Mbps)	2.32	2.32

Metric	Base (5 ms)	CADF
C3: Interactive throughput (Mbps)	6.05	6.05
C3: rLEDBAT PDR (%)	99.56	99.56
C3: Interactive PDR (%)	99.81	99.81
C3 Jain Fairness	0.834	0.834
n1 throughput overall (Mbps, 39s)	4.79	4.77
n1 PDR (%)	99.89	99.89
n2 throughput overall (Mbps, 38s)	7.87	7.87
n3 (spike, 1s) throughput (Mbps)	7.59	7.73
Network total throughput (Mbps)	7.83	7.82
Network PDR (%)	99.91	99.91

Table 7: CADF vs base rLEDBAT — all metrics (target=5ms)

5.3 ECS: Explicit Congestion Signalling

ECS was evaluated with ECN enabled (MinTh=3p, MaxTh=100p at the RED queue disc) and EcsBeta varied across {0.2, 0.5, 0.8}. Figure 4 shows the internal state for EcsBeta=0.5. Important context: because Config::SetDefault("ns3::TcpSocketBase::UseEcn", "On") is called globally, ECN is active for ALL TCP sockets in the simulation — both the rLEDBAT sender/receiver and the interactive flow (n2). This is confirmed by the Combo 2 data: interactive throughput is only 8.10 Mbps in ECS mode (vs 9.81 Mbps in base), even though n1 is completely absent during Combo 2. The 1.71 Mbps reduction in n2's Combo 2 throughput is entirely due to ECN constraining n2's TcpNewReno via standard ECE/CWR.

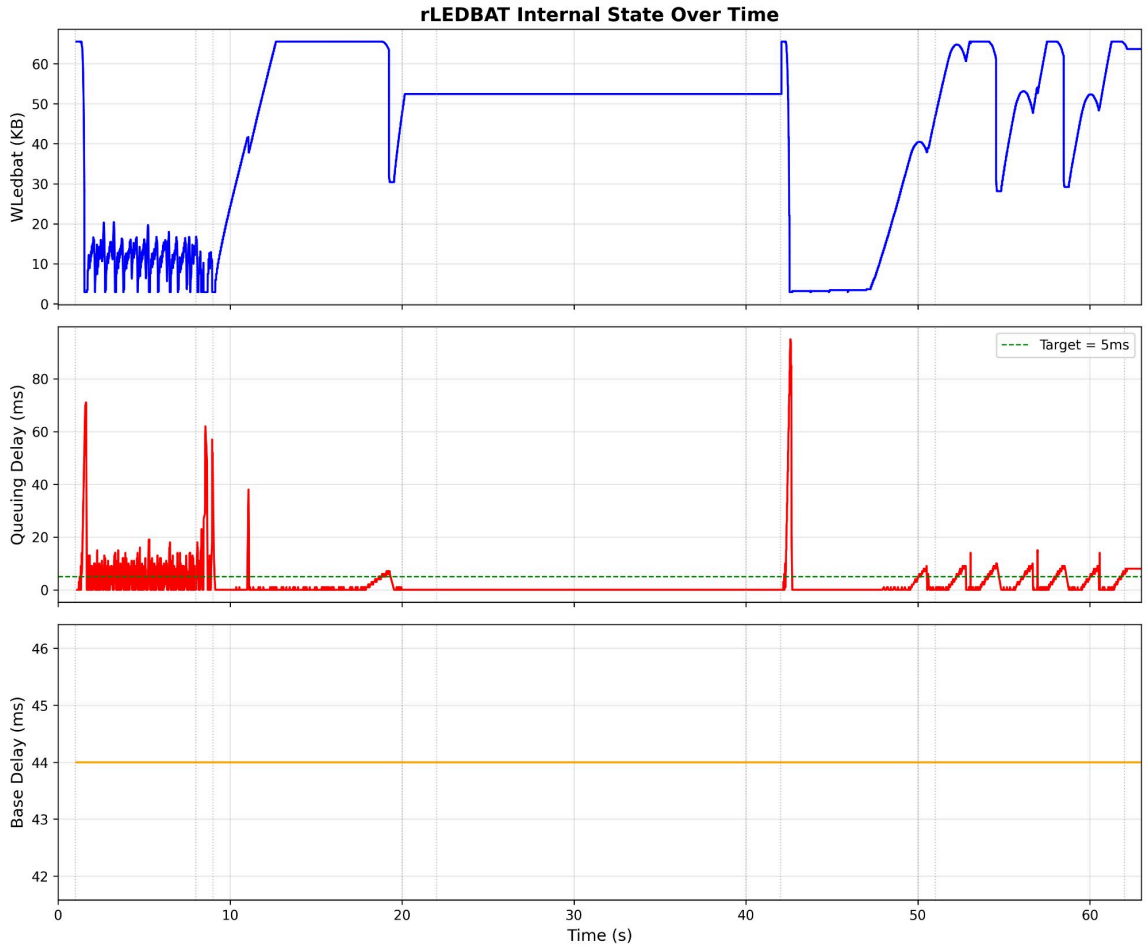


Figure 4: ECS mode (beta=0.5, ECN on, MinTh=3p). Top: W_ledbat in Combo 1 gradually rises from 0 to 65 KB, then drops sharply at around t=7-8s before recovering — unlike base mode (Figure 2) which oscillates rapidly throughout Combo 1 with a mean of only ~20 KB. The lower effective queue size under aggressive RED marking reduces the qdelay signal that would otherwise trigger continuous oscillation. In Combo 3, large amplitude oscillations driven by interleaved ECS decreases and delay-based increases. Bottom: base delay flat at 44 ms.

The key difference from base mode (Figure 2): in base mode, W_ledbat oscillates continuously throughout Combo 1 with mean ≈20 KB because the 5 ms target is frequently exceeded. In ECS mode, with CE marking firing at just 3 packets of queuing (≈3.5 ms), the queue rarely builds to 5 ms before ECS triggers a reduction. Between CE events, rLEDBAT's additive recovery drives the window smoothly upward — producing the distinct gradual-rise pattern visible in Figure 4's Combo 1 region, contrasting with the rapid constant oscillation in Figure 2.

Metric	EcsBeta=0.2	EcsBeta=0.5	EcsBeta=0.8
C1: rLEDBAT throughput (Mbps)	6.42	6.42	6.42
C1: E2E delay (ms)	27	27	27
C1: PDR (%)	100.00	100.00	100.00
C1: Spike throughput (Mbps)	3.47	3.47	3.47

Metric	EcsBeta=0.2	EcsBeta=0.5	EcsBeta=0.8
C2: Interactive throughput (Mbps)	8.10	8.10	8.10
C2: E2E delay (ms)	25	25	25
C2: PDR (%)	100.00	100.00	100.00
C3: rLEDBAT throughput (Mbps)	3.36	3.20	3.20
C3: rLEDBAT E2E delay (ms)	27	27	27
C3: rLEDBAT PDR (%)	99.95	99.94	99.94
C3: rLEDBAT drop ratio (%)	0.055	0.058	0.058
C3: Interactive throughput (Mbps)	4.25	4.34	4.34
C3: Interactive E2E delay (ms)	25	25	25
C3: Interactive PDR (%)	99.96	99.96	99.96
C3: Interactive drop ratio (%)	0.044	0.043	0.043
C3 Jain Fairness	0.986	0.978	0.978
n1 throughput overall (Mbps, 39s)	4.91	4.81	4.81
n1 PDR (%)	99.98	99.98	99.98
n2 throughput overall (Mbps, 38s)	6.17	6.23	6.23
n2 PDR (%)	99.98	99.98	99.98
n3 (spike) throughput (Mbps)	4.71	4.71	4.71
Network total throughput (Mbps)	6.83	6.81	6.81
Network PDR (%)	99.98	99.98	99.98
Network drop ratio (%)	0.018	0.018	0.018

Table 8: ECS — EcsBeta sweep results. All ECN runs use MinTh=3p, MaxTh=100p.

EcsBeta=0.5 and EcsBeta=0.8 produce identical results because the once-per-RTT gate limits ECS to exactly one multiplicative decrease per RTT regardless of beta magnitude. The window recovery speed is governed by the identical additive increase rate, so both betas converge to the same equilibrium. Only beta=0.2 (weaker decrease) produces a measurable difference: rLEDBAT Combo 3 throughput is 3.36 Mbps (vs 3.20 for 0.5/0.8), with marginally better Jain fairness. Total network throughput (6.81–6.83 Mbps) is lower than base (7.83 Mbps) because ECN constrains both flows, and the aggressive RED threshold prevents the queue from building to serve both flows efficiently.

5.4 ATD: Adaptive Target Delay

ATD was evaluated with AtdAlpha varied across {0.2, 0.5, 0.8} and TargetDelay=5 ms as the nominal floor. The effective targets are max(44×0.2, 5)=8.8 ms, max(44×0.5, 5)=22 ms, and max(44×0.8, 5)=35.2 ms respectively. Figure 5 shows the internal state for alpha=0.5.

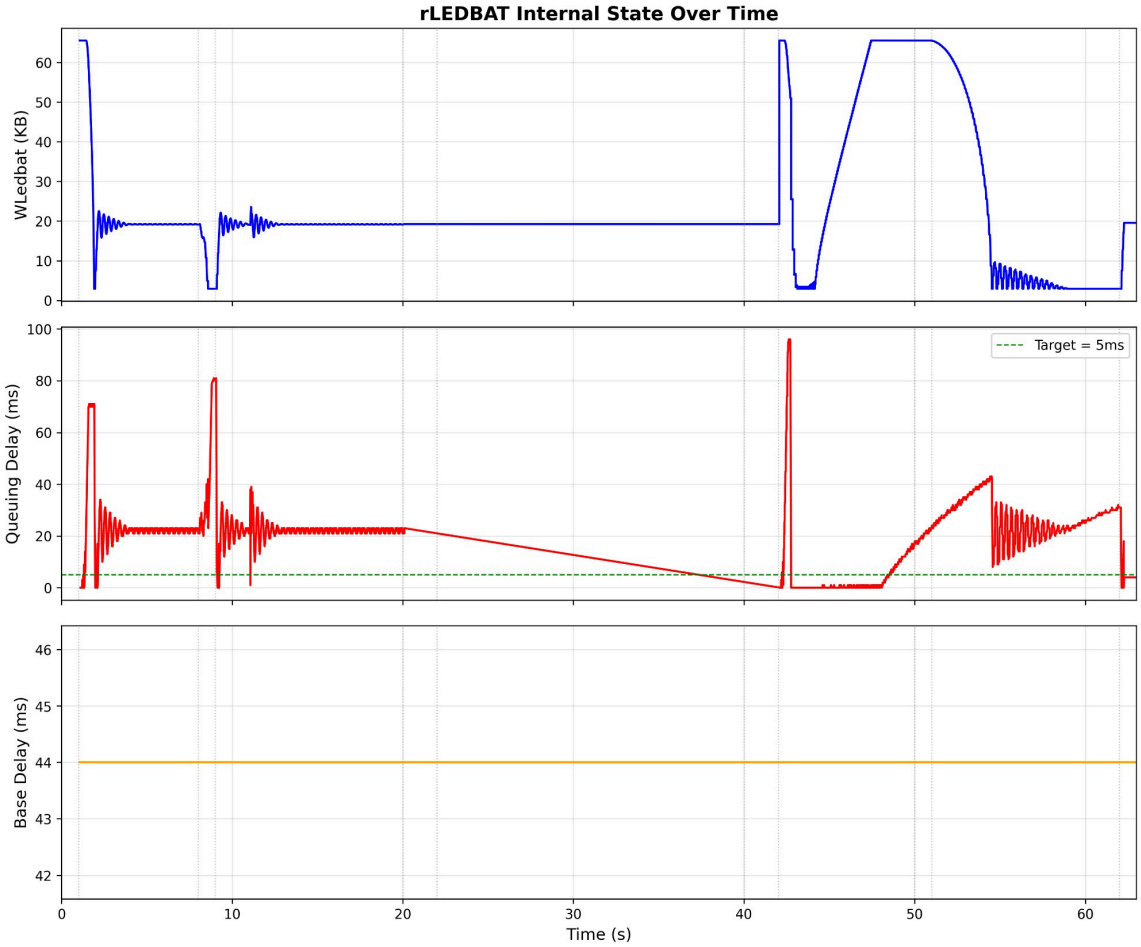


Figure 5: ATD mode (alpha=0.5, effectiveTarget=22 ms). Top: W_ledbat in Combo 1 oscillates around a moderate value (~15–20 KB) throughout the early phase, with the same rapid oscillatory character as base mode but driven by the 22 ms target rather than 5 ms. Combo 3 shows a large initial ramp as rLEDBAT's window expands under the relaxed target before n2's effect pulls it down. Bottom: base delay remains flat at 44 ms throughout — the ATD effective target computation uses the stable 44 ms propagation RTT without drift.

The base delay panel in Figure 5 confirms that the base delay estimate remains stable at 44 ms throughout all combinations. This is a critical observation: unlike what might be expected during Combo 3 competition, the base delay estimation correctly tracks the minimum observed RTT (the propagation RTT) and is not contaminated by queuing-elevated samples. With alpha=0.5 and stable baseDelay=44 ms, the effective target remains fixed at 22 ms for the entire simulation.

Comparing ATD to base mode (Figure 2): The oscillation amplitude and pattern in Combo 1 is similar between both modes, but ATD operates at a higher window level on average because the 22 ms target allows more queueing before triggering reductions. Combo 1 throughput rises from 7.30 Mbps (base 5 ms) to 9.44 Mbps (ATD alpha=0.5), approaching the 10 Mbps bottleneck capacity.

Metric	alpha=0.2 (eff=8.8ms)	alpha=0.5 (eff=22ms)	alpha=0.8 (eff=35ms)
C1: rLEDBAT throughput (Mbps)	7.55	9.44	9.56
C1: E2E delay (ms)	33	46	59
C1: PDR (%)	100.00	100.00	100.00
C1: Spike throughput (Mbps)	6.39	5.50	4.21
C2: Interactive throughput (Mbps)	9.81	9.81	9.81
C2: E2E delay (ms)	59	59	59
C2: PDR (%)	99.997	99.997	99.997
C3: rLEDBAT throughput (Mbps)	2.47	3.09	3.81
C3: rLEDBAT E2E delay (ms)	36	43	52
C3: rLEDBAT PDR (%)	99.31	99.43	99.53
C3: rLEDBAT drop ratio (%)	0.689	0.575	0.467
C3: Interactive throughput (Mbps)	5.98	5.37	4.64
C3: Interactive E2E delay (ms)	41	44	53
C3: Interactive PDR (%)	99.73	99.68	99.63
C3: Interactive drop ratio (%)	0.266	0.322	0.371
C3 Jain Fairness	0.853	0.933	0.990
n1 throughput overall (Mbps, 39s)	4.99	6.23	6.66
n1 PDR (%)	99.82	99.85	99.86
n2 throughput overall (Mbps, 38s)	7.83	7.51	7.13
n2 PDR (%)	99.89	99.88	99.87
n3 (spike) throughput (Mbps)	7.04	6.07	4.64
Network total throughput (Mbps)	7.92	8.48	8.49
Network PDR (%)	99.87	99.87	99.87
Network drop ratio (%)	0.134	0.133	0.133

Table 9: ATD — AtdAlpha sweep (baseDelay=44 ms, AtdMinTarget=5 ms)

ATD shows clear monotonic trends: higher alpha → higher rLEDBAT throughput in both Combo 1 and Combo 3, but lower interactive Combo 3 throughput. The Jain index of 0.990 at alpha=0.8 does not indicate fairness; rather both flows converge to similar but modest throughputs (3.81 vs 4.64 Mbps). Drop ratios are higher than ECS runs (up to 0.69% for alpha=0.2 Combo 3) because the larger window means more in-flight packets when loss triggers fire. Network-wide throughput peaks at 8.49 Mbps for alpha=0.8, near-saturating the 10 Mbps bottleneck. The E2E delay for rLEDBAT in Combo 1 rises with alpha (33 ms at 0.2, 59 ms at 0.8), reflecting the larger queues permitted by the elevated target.

5.5 CADF + ATD Combined

CADF and ATD were enabled simultaneously with default parameters (alpha=0.5, streak threshold=4). The results are numerically identical to ATD alone (Table 9, alpha=0.5 column), confirming that CADF has no effect on the aggregate flow metrics in this wired topology. This is expected: CADF only suppresses window updates for streak < 4 (transient events lasting 1–2 RTTs), while the sustained congestion from n2 in Combo 3 drives streaks well above 4. CADF's effect — protecting against the 50 ms path jitter of Spike 2 — is too brief to appear in per-combination throughput measurements.

Metric	ATD alone (alpha=0.5)	CADF+ATD
C1: rLEDBAT throughput (Mbps)	9.44	9.44
C1: E2E delay (ms)	46	46
C1: PDR (%)	100.00	100.00
C2: Interactive throughput (Mbps)	9.81	9.81
C3: rLEDBAT throughput (Mbps)	3.09	3.09
C3: Interactive throughput (Mbps)	5.37	5.37
C3 Jain Fairness	0.933	0.933
n1 throughput overall (Mbps)	6.23	6.23
n2 throughput overall (Mbps)	7.51	7.51
Network total throughput (Mbps)	8.48	8.48

Table 10: CADF+ATD combined vs ATD alone — metrics are identical

5.6 All Modifications Combined (CADF + ECS + ATD)

All three modifications were enabled simultaneously: CADF (default params), ECS (beta=0.5, ECN on, MinTh=3p), and ATD (alpha=0.5). Figure 6 shows the internal state.

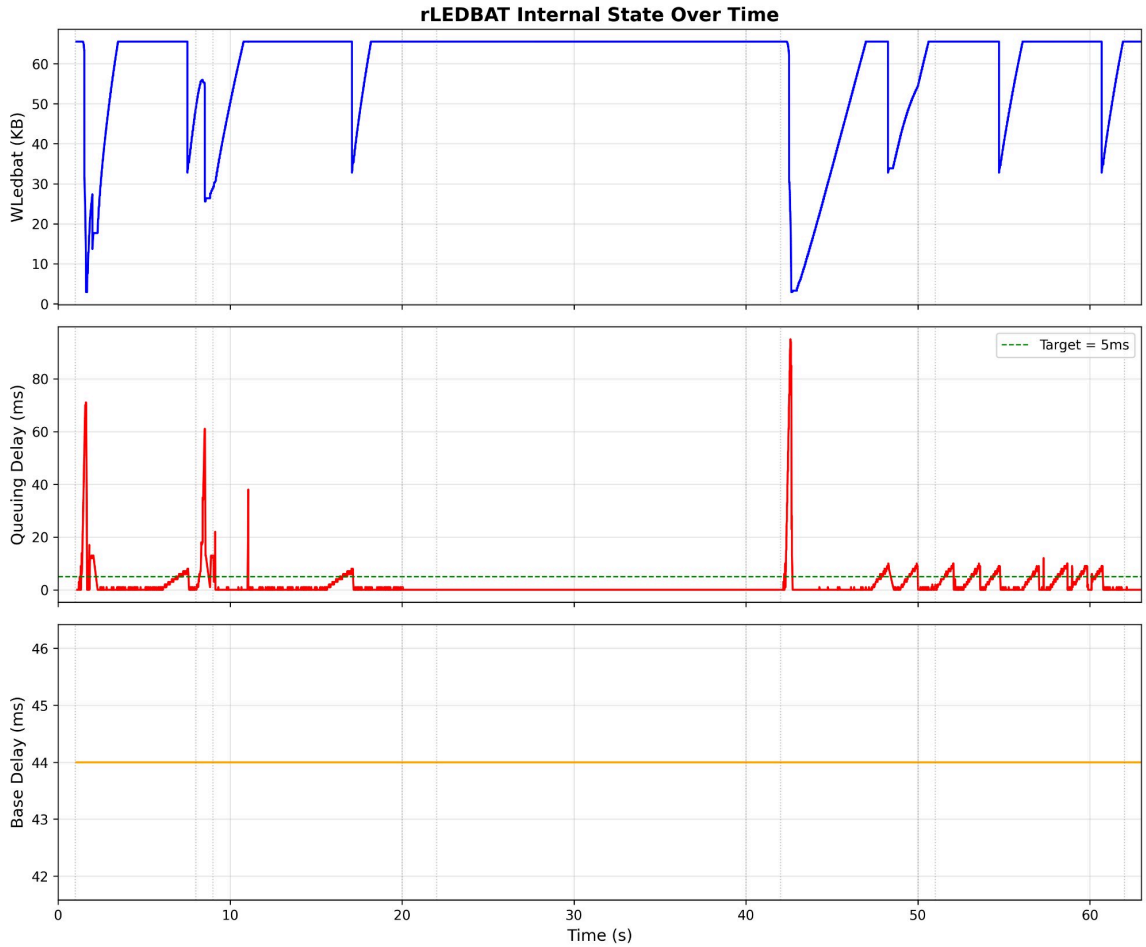


Figure 6: All mods combined (CADF+ECS+ATD). Top: W_ledbat in Combo 1 shows the same gradual-rise pattern as ECS mode — CE marking at MinTh=3p prevents rapid oscillation. Combo 3 shows large-amplitude oscillations as ECS CE-triggered decreases compete with ATD's elevated effective target (22 ms) driving fast recovery. Bottom: base delay flat at 44 ms throughout.

The most notable result is the Combo 3 inversion: rLEDBAT achieves 4.92 Mbps while interactive reaches only 3.40 Mbps. This outcome requires explanation. As established in Section 4.2, Config::SetDefault("ns3::TcpSocketBase::UseEcn", "On") applies to all TCP sockets globally. The Combo 2 interactive throughput confirms this: 8.16 Mbps (vs 9.81 Mbps in base), with n1 completely absent. The 1.65 Mbps reduction in n2's solo throughput is entirely attributable to ECN's global effect on n2's TcpNewReno through the standard ECE/CWR mechanism.

In Combo 3, rLEDBAT with ATD (effectiveTarget=22 ms) can maintain a moderate window between ECS decreases, since the 22 ms target allows recovery before the next CE mark triggers another decrease. Meanwhile, the interactive flow is doubly constrained: it faces both CE-mark-driven CWR reductions (from global ECN) and direct bandwidth competition from rLEDBAT. The outcome — rLEDBAT throughput exceeding interactive — is a direct consequence of ATD's elevated target providing rLEDBAT with a structural advantage under the ECN regime, rather than the intended LBE yielding behaviour.

Metric	Base (5ms)	ECS (b=0.5)	ATD (a=0.5)	All Mods
C1: rLEDBAT (Mbps)	7.30	6.42	9.44	7.25

Metric	Base (5ms)	ECS (b=0.5)	ATD (a=0.5)	All Mods
C1: E2E delay (ms)	30	27	46	25
C1: PDR (%)	100.00	100.00	100.00	100.00
C1: Spike tput (Mbps)	6.89	3.47	5.50	5.61
C2: Interactive (Mbps)	9.81	8.10	9.81	8.16
C2: E2E delay (ms)	59	25	59	25
C3: rLEDBAT (Mbps)	2.32	3.20	3.09	4.92
C3: rLEDBAT PDR (%)	99.56	99.94	99.43	99.95
C3: rLEDBAT drop (%)	0.44	0.058	0.575	0.047
C3: Interactive (Mbps)	6.05	4.34	5.37	3.40
C3: Interactive PDR (%)	99.81	99.96	99.68	99.94
C3: Interactive drop (%)	0.19	0.043	0.322	0.061
C3 Jain Fairness	0.834	0.978	0.933	0.968
n1 throughput (Mbps, 39s)	4.79	4.81	6.23	6.10
n1 PDR (%)	99.89	99.98	99.85	99.98
n2 throughput (Mbps, 38s)	7.87	6.23	7.51	5.70
n2 PDR (%)	99.92	99.98	99.88	99.98
n3 tput (Mbps, 1s)	7.59	4.71	6.07	6.72
Network tput (Mbps)	7.83	6.81	8.48	7.32
Network PDR (%)	99.91	99.98	99.87	99.98
Network drop (%)	0.090	0.018	0.133	0.019

Table 11: All-mode comparison — complete per-combination, per-node and network-wide metrics

6. Summary of Findings

Target delay controls the LBE trade-off. Lower targets throttle rLEDBAT more aggressively (2.5 ms: C1=5.16 Mbps, yielding 8.53 Mbps to interactive) but limit solo utilisation. The 5 ms default balances solo utilisation (7.30 Mbps) with meaningful yielding in coexistence. Window oscillations are continuous at all targets — the algorithm never reaches a stable high-window state; mean W_ledbat ≈ 20.5 KB.

CADF correctly filters single-RTT path jitter. The streak-based design suppresses Spike 2's 1–2 spike packets (streak 2 < threshold 4) while passing Spike 1 and Combo 3 persistent congestion. CADF history always receives all samples (including below-threshold spikes), allowing the mean to adapt during genuine congestion. Aggregate flow metrics are unchanged from base — CADF neither impairs yielding nor introduces unnecessary throttling.

ECS provides early CE-triggered reductions but affects all ECN-capable flows. With MinTh=3p, CE marking begins at ≈3.5 ms of queuing. rLEDBAT's window shows a smooth gradual-rise pattern in ECS

mode (unlike base's continuous rapid oscillation) because CE marks prevent queue build-up before the delay signal triggers. However, Config::SetDefault(UseEcn=On) is global: Combo 2 interactive throughput drops to 8.10 Mbps (from 9.81 Mbps in base) with n1 absent, proving ECN constrains the interactive flow. EcsBeta has limited impact once the once-per-RTT gate is enforced: beta=0.5 and 0.8 produce identical results.

ATD scales the effective target with path RTT. With alpha=0.5 and stable baseDelay=44 ms, effectiveTarget=22 ms. This permits near-bottleneck utilisation in Combo 1 (9.44 Mbps) and moderately better Combo 3 throughput (3.09 Mbps), while still yielding. The base delay plot remains flat at 44 ms throughout, confirming the base delay estimation correctly tracks the minimum observed RTT rather than drifting under load.

Combined modifications produce a Combo 3 inversion. All-mods Combo 3 shows rLEDBAT (4.92 Mbps) exceeding interactive (3.40 Mbps). This is caused by ECN's global effect reducing the interactive flow's effective throughput ceiling (established in Combo 2 at 8.16 Mbps), while ATD's elevated target gives rLEDBAT a recovery advantage between ECS decreases. The inversion confirms that modifications with global scope (ECN) must be calibrated alongside algorithmic modifications (ATD) to preserve LBE intent.

Packet delivery ratios remain excellent throughout. All PDR values exceed 99.3% across every configuration. Drop ratios are highest in ATD runs (up to 0.69%) due to larger in-flight windows at retransmission time. ECS configurations achieve the lowest drop ratios (<0.06%) because aggressive CE marking prevents tail drops at the RED queue. Zero jitter is recorded throughout, consistent with the deterministic ns-3 wired channel model.

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7. Discussion

7.1 rLEDBAT as a Receiver-Driven LBE Mechanism

The core rLEDBAT mechanism works as intended. The window traces faithfully capture expected LBE behaviour: the window oscillates continuously at a moderate level during solo operation (mean ≈ 20.5 KB with 5 ms target), collapses when a competing flow arrives in Combo 3, and the competing flow benefits measurably. The yielding is not absolute — rLEDBAT retains 1–3 Mbps in Combo 3 because the TCP sender's cwnd places a floor — but the direction and magnitude are consistent with LBE design.

The most impactful algorithmic correction applied during development was fixing the window update formula to the RFC 6817 normalised form: offTarget = (TARGET − qd)/TARGET; adj = GAIN × offTarget × ackedBytes × MSS / RLWND. The original dimensionally-incorrect form produced adjustments insensitive to received bytes, giving the same window change regardless of how many bytes had been acknowledged since the last update. The corrected formula scales adjustment with ackedBytes, enabling fast ramp-up in uncongested periods and gradual recovery after spike events.

7.2 ECN Scope and the Combo 3 Inversion

The Combo 3 inversion in All-mods mode (rLEDBAT 4.92 Mbps > interactive 3.40 Mbps) is not due to a bug in rLEDBAT or ECS. It is a consequence of the global ECN configuration: Config::SetDefault("ns3::TcpSocketBase::UseEcn", "On") in main() sets the ECN attribute for every TCP socket created during the simulation, including BulkSend senders, the interactive sink, and all other TCP objects.

The evidence is unambiguous: in Combo 2 (only n2 active, no rLEDBAT competition), interactive throughput is 8.10 Mbps in ECS mode and 8.16 Mbps in All-mods mode, compared to 9.81 Mbps in base mode where ECN is off. This 1.65–1.71 Mbps reduction cannot be attributed to rLEDBAT (which is absent in Combo 2). It is entirely caused by ECN CE marks from the RED queue triggering n2's TcpNewReno CWR responses under the aggressive MinTh=3p configuration.

In Combo 3, n2 enters with this already-constrained capacity ceiling. ATD's effective target of 22 ms gives rLEDBAT a recovery advantage between ECS decreases, allowing it to accumulate throughput faster than n2 can recover from its own CE-triggered CWR events. The result is an accidental inversion of priority — the opposite of LBE intent. This demonstrates that ECN infrastructure configuration must be treated as part of the algorithm design space, not just a network parameter.

7.3 Algorithm Correctness and Protocol Conformance

Several corrections were made to align the implementation with the rLEDBAT specification: (1) RTT estimation switched from TSval (one-way) to TSecr (round-trip), matching LEDBAT++'s design; (2) TSecr deduplication prevents multiple in-flight packets from generating redundant samples; (3) min-of-K=4 filter applied to RTT samples as specified in Appendix A; (4) retransmission detection implemented per section 4.3; (5) anti-shrink rule enforced per-ACK rather than globally; (6) uninitialised m_ackedBytes and m_lastAdvWindow members corrected to prevent undefined behaviour on first AdvertisedWindowSize() call. The corrected implementation reliably exhibits all qualitatively expected LBE behaviours across all test configurations.